NORTH CAROLINA AGRICULTURAL AND TECHNICAL STATE UNIVERSITY

MATH 224- Data Analysis Project

Part II: Further Data Exploration and Inference

AI Tool Usage Instructions

For this project, you are allowed to use one generative AI tool of your choice. Examples include ChatGPT, Microsoft Copilot, or Google Gemini. Our school currently provides free access to Microsoft Copilot, and Gemini offers a free student plan for 12 months — but it is completely up to you which tool you prefer.

Rules for AI Use

- Use only ONE dedicated conversation/chat for this project. All project-related prompts and questions must stay in that same conversation.
- Do not spread project-related questions across multiple chats. If you need help for unrelated study purposes, open a different chat.
- Include your AI interaction link upon submission so your instructor can review your full conversation.
- AI is a tool to assist your thinking — not a replacement for it. You must understand, verify, and take ownership of every answer you submit.

What You SHOULD Do ✓

✓ Good AI Use — Examples

"I have a dataset with variables age, income, and education. Can you explain what summary statistics I should compute and why?"

"My histogram looks right-skewed. What does that mean for interpreting the mean vs. median?"

"Here is my R code — it gives an error. Can you help me find the issue?" [paste your actual code]

"Can you explain what a confidence interval means in simple terms?"

"I got a p-value of 0.03 in my t-test. How do I interpret this in context?"

What You Should NOT Do ✗

✗ Prohibited AI Use — Do NOT do these

Do NOT paste the project question directly and submit whatever the AI writes.
Do NOT ask: "Do Part II of my MATH 224 project for me."
Do NOT copy and paste AI output word-for-word without understanding it.
Do NOT use AI to generate R code you cannot explain or did not verify.
Do NOT use multiple chat threads to hide the extent of your AI use.

⚠ Important Reminder

AI tools can make mistakes. Always verify outputs against your course notes and textbook, and ask your instructor.

Reflection Requirement

At the end of your submission, include a short paragraph (5–8 sentences) reflecting on how you used the AI tool. Address the following:

- Which AI tool did you use and for which tasks?
- Did the AI make any mistakes? How did you catch and correct them?
- How did using AI affect your understanding of the material?
- What would you do differently next time?

General Instructions

Use the R Markdown report outline template you created in RStudio Cloud during lab. Knit your R Markdown code as a PDF and submit it — including all code and answers — under "Data Analysis Project – Part II" in Blackboard by the due date.

In Part I, you explored your dataset, identified research questions, and performed exploratory data analysis. In this part, you will extend that work with statistical inference.

Project Tasks

Task 1 — Repeat Summary Statistics (5 points)

Repeat Task 2d from Part I: compute and report summary statistics (mean, standard deviation, and five-number summary) for the response variable you identified in Part I.

💡 AI Prompt Idea

"My summary statistics from Part I showed [values]. My new output shows [values]. Can you help me understand if there is any meaningful change?"

Task 2 — Confidence Interval for Population Mean (10 points)

2a. Check Assumptions (2.5 points)

Check the assumptions needed to construct a confidence interval for the population mean. Make a histogram of your response variable and check if it is approximately normally distributed. Report your sample size and explain why the Central Limit Theorem applies.

> 💡 **AI Prompt Idea for 2a**
>
> *"My histogram of [variable] looks [shape]. Does this satisfy the normality assumption for constructing a confidence interval? My sample size is [n]."*
>
> *"Can you explain in simple terms when the Central Limit Theorem allows us to use a t-interval even if the data is not perfectly normal?"*

2b. Compute 95% Confidence Interval (5 points)

Use the R function t_test() to compute the 95% confidence interval for the population mean of your response variable. Show your code and output.

> 💡 **AI Prompt Idea for 2b**
>
> *"I am using t_test() to compute a confidence interval. Here is my code: [paste code]. Can you check if I am using the function correctly?"*

> ✗ **Do NOT do this**
>
> Do NOT ask AI to generate the R code and paste it without running or understanding it.
>
> Do NOT report a confidence interval without also showing the R code that produced it.

2c. Interpret the Confidence Interval (2.5 points)

Interpret the 95% confidence interval you obtained in the context of your dataset.

> 💡 **AI Prompt Idea for 2c**
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> *"My 95% confidence interval for mean income is ($48,000, $54,000). Can you help me write a correct interpretation in plain language?"*

Task 3 — Repeat Association Graphs (4 points)

Repeat Task 2f from Part I: create at least two graphs showing the association between your response variable and two explanatory variables (one using a scatter plot for a quantitative explanatory variable, and one using a side-by-side boxplot for a categorical explanatory variable).

Task 4 — Group Summary Statistics (4 points)

Compute the mean and standard deviation of your response variable by the levels of one categorical variable. For example, compare the mean body weight for men vs. women. Comment on the patterns you observe.

AI Prompt Idea for Task 4

"I grouped my data by [categorical variable] and got these means: [paste table]. What patterns should I look for and how do I comment on them statistically?"

Task 5 — Confidence Interval for Difference in Means (7 points)

5a. Compute 90% Confidence Interval for Difference (5 points)

Use t_test() to compute and report the 90% confidence interval for the difference in population means across the levels of one categorical explanatory variable. Show your code and output.

AI Prompt Idea for 5a

"I want to compute a 90% confidence interval for the difference in mean income between men and women using t_test() in R. Here is my dataset structure: [describe or paste glimpse() output]. What arguments do I need?"

5b. Interpret the Confidence Interval (2 points)

Interpret the confidence interval you obtained in the context of your dataset.

AI Prompt Idea for 5b

"My 90% Confidence Interval for the difference in mean income between men and women is ($3,200, $8,900). Can you help me write an interpretation and explain what it means that zero is not in the interval?"

Submission Checklist

- R Markdown knitted as a PDF with all code visible
- Copy and paste the entire AI conversation, in addition to the link, into a text doc (.txt)
